

# Olympiad Test Paper — Science (Class 7) — Paper 5

## Chapter 16: Water: A Precious Resource

### Paper 5 — (progressively harder)

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| <b>Subject</b>         | Science (NCERT Class 7)              |
| <b>Chapter</b>         | 16 — Water: A Precious Resource      |
| <b>Total Questions</b> | 60 (Multiple Choice, single correct) |
| <b>Maximum Marks</b>   | 240                                  |
| <b>Suggested Time</b>  | 90 minutes                           |

**Marking scheme:** +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

**Instructions:** Each question has four options (A), (B), (C), (D); exactly one is correct. Every distractor is a hand-built expert trap — surface coverage read as usable supply, recharge rate confused with stored volume, a falling table's symptoms mistaken for its cause, multi-year balance arithmetic inverted, drip and harvesting effects reversed, transpiration credited as condensation, the cycle's constant total misread as creation or net loss, and state changes named in the wrong order. Do not write on this paper — mark answers on your answer sheet.

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**1.** A single labelled water molecule starts in the ocean, evaporates, joins a cloud, falls as snow on a glacier, melts decades later into a river, and finally flows back to the sea. Counting only changes of physical state along this whole journey, how many state changes did it undergo, and what does that confirm about the chapter's claim that Earth's total water stays roughly constant?

- (A) Four state changes, and each change adds a little new water to the system, slowly increasing Earth's total over the centuries
- (B) Two state changes, and the snowfall step destroyed some of the water, which is why the total slowly falls
- (C) Three state changes, but because the molecule travelled so far, an equal mass of water must have been lost to space to keep the total constant
- (D) Three state changes (liquid to vapour, vapour to solid, solid to liquid); the same water merely changed form and place repeatedly without any being created or destroyed, which is exactly why the global total holds steady

**2.** Inside a sealed terrarium left in sunlight, soil moisture vanishes from the earth, mist beads on the cool glass roof, and droplets later trickle back to the soil — yet the jar's sealed

mass never changes. The pupil who best understands the chapter would say this miniature system proves that:

- (A) the plants are slowly creating extra water through their leaves, which is why the soil keeps getting re-wetted
- (B) a closed water cycle can run on its own with no net gain or loss — evaporation, condensation and a rain-like return simply recirculate the same fixed amount of water
- (C) water is being lost each cycle through the glass, so the soil will eventually dry out for good
- (D) the droplets on the glass are a new substance formed by sunlight, not the same water that left the soil

**3.** On a freezing dawn a metal railing carries white frost; by midday the frost is clear water beads; by dusk the railing is bone dry; overnight the rail again grows frost. Naming the state changes strictly in the order they happened across this full day-and-night, the sequence is:

- (A) evaporation first, then melting, then condensation into the morning frost
- (B) melting (solid to liquid), then evaporation (liquid to vapour), then the vapour later turning straight back to solid frost (deposition-like return to ice)
- (C) freezing, then condensation, then melting in that single order
- (D) condensation of vapour into frost, then evaporation, then melting of the beads

**4.** A child claims the visible white plume above a boiling pot and the invisible dampness that later mists a cold tile are two unrelated things, and adds that the plume is 'pure gas'. The chapter-accurate correction is that:

- (A) both are water, and the child is right that the visible white plume is the true gas while the invisible dampness is liquid
- (B) the plume is water but the invisible dampness is ordinary air, a completely different substance
- (C) they are two different chemicals that only behave alike because both are hot
- (D) both are the same substance, water; the white plume is already tiny condensed droplets (liquid), while the invisible dampness is the true gaseous vapour — so it is the invisible part, not the plume, that is the real gas

**5.** Within one frozen lake on a winter afternoon, a skater stands on the ice sheet, fish swim in the liquid water below it, and breath-fog hangs in the air above. Asked what single principle this scene best demonstrates, the strongest answer is that:

- (A) water must adopt only a single state across any one body of water at any given time
- (B) all three physical states of the one substance water can be present at the same moment when conditions vary from one spot to the next
- (C) each of the three forms here is a chemically different material that merely resembles water
- (D) ice can occur only at the polar regions and never directly above liquid water

**6.** A demonstrator cools a sealed flask of humid air so quickly that droplets form on its walls without the air ever rising or moving. A student protests, 'But clouds need air to rise to form!' The most precise resolution is:

- (A) the student is right; without rising air no condensation can occur, so the flask droplets must have a different cause
- (B) rising air is just the usual way the air gets cool enough; the actual trigger for droplets, here and in a cloud, is cooling the vapour until it condenses — achieved in the flask directly by chilling
- (C) the droplets formed because the vapour was heated, which is what truly makes clouds as well
- (D) the flask droplets are frozen ice, not condensation, so the rising-air rule still stands unbroken

**7.** Over one summer day a thick mango orchard releases far more moisture into the air than an equal-area gravel car park beside it. A pupil credits this to 'condensation, since shady leaves are cooler.' The chapter-correct explanation is that the extra moisture comes from:

- (A) condensation on the cool leaves, exactly as the pupil reasoned, since cooler surfaces always add water to the air
- (B) transpiration — the trees draw up soil water through their roots and release it as vapour from their leaves, adding moisture the bare car park cannot
- (C) infiltration, because forest soil forces stored water upward and out into the air
- (D) precipitation generated privately above the orchard canopy and nowhere else

**8.** Arrange these four events into the single correct order they occur for one parcel of sea water completing exactly one full loop of the water cycle: (i) it seeps through soil toward the groundwater, (ii) it falls on a hill as rain, (iii) it condenses into a cloud, (iv) it rises off the sea as vapour.

- (A) condenses into a cloud, rises as vapour, seeps through the soil, then falls as rain
- (B) falls as rain, rises as vapour, condenses into a cloud, then seeps through the soil
- (C) seeps through the soil, falls as rain, condenses into a cloud, then rises as vapour
- (D) rises off the sea as vapour, condenses into a cloud, falls on the hill as rain, then seeps through soil toward the groundwater

**9.** A teacher writes 'sleet, drizzle, hail and snow' on the board and asks why all four sit under one stage of the water cycle, even though one is solid, others liquid, and they fall at very different speeds. The unifying reason is that each is a way that:

- (A) water returns from the air to the ground, so all four are forms of precipitation regardless of their state or speed
- (B) water leaves the ground and enters the air, so all four count as evaporation
- (C) vapour is condensing high in the sky, so all four are best classed as condensation
- (D) plants are giving off moisture, so all four are kinds of transpiration

**10.** Two posters hang side by side. Poster X: 'Earth's surface is ~71% water — supply is huge.' Poster Y: '~97% of all water is ocean.' A student blends them into 'so about 71% of Earth's water is fresh and drinkable.' The single deepest flaw is that the student has:

- (A) correctly reasoned that 71% surface coverage must mean 71% of the water is fresh and ready to drink
- (B) wrongly trusted the 97% ocean figure, since oceans are in fact fresh and fully drinkable
- (C) confused how much of the surface is covered by water with how much of the total water is usable fresh water — and ignored that ~97% being ocean leaves only a tiny fresh, mostly locked-away fraction
- (D) merely overestimated by a few per cent; about 71% of Earth's water really is drinkable fresh water

**11.** Imagine all of Earth's water poured into exactly 1000 identical jugs. About 970 hold salty ocean water. Of the remaining ~30 fresh-water jugs, most are frozen in ice and glaciers. Which statement about the jugs is most accurate within the chapter?

- (A) the vast majority of the 1000 jugs are ocean, only about 30 are fresh, and even within those 30 most are locked as ice — so the freely usable share is a small slice of an already small fraction
- (B) about 710 of the 1000 jugs are fresh and drinkable, matching the 71% surface figure
- (C) roughly 500 jugs are salty and 500 are fresh, an even split of Earth's water
- (D) about 970 of the jugs are fresh and only 30 are salty ocean water

**12.** A tour guide says: 'Glaciers store the lion's share of Earth's fresh water, so glacier ice must be where our towns' daily tap water mainly comes from.' Pin down the exact half that is right and the half that is wrong:

- (A) both halves are right: glaciers hold most fresh water and towns indeed draw their daily tap water straight from melting glacier ice
- (B) both halves are wrong: glaciers hold almost no fresh water and are irrelevant to supply
- (C) the first half is right — glaciers do hold most fresh water — but the second is wrong, because that water is locked as ice, so everyday supply comes mainly from groundwater and surface sources
- (D) the first half is wrong but the second is right: glaciers hold little fresh water, yet daily tap water still comes from glaciers

**13.** Rank these four from the LARGEST store of water to the smallest, using chapter knowledge: the oceans; groundwater; the water frozen in all ice-caps and glaciers; the water flowing in all the world's rivers at this instant.

- (A) rivers' flowing water, then groundwater, then ice-caps and glaciers, then oceans
- (B) oceans, then ice-caps and glaciers, then groundwater, then the rivers' flowing water

- (C) oceans, then groundwater, then rivers' flowing water, then ice-caps and glaciers
- (D) ice-caps and glaciers, then oceans, then rivers' flowing water, then groundwater

**14.** Two neighbouring districts in one state receive identical annual rainfall, yet over a decade District A enjoys steady wells while District B faces chronic shortage. The conclusion most faithful to the chapter is that fresh-water availability:

- (A) is distributed unevenly because it depends not only on rainfall but on how much is stored, how much is pumped out, and how much rain is allowed to recharge in each place
- (B) must be identical wherever rainfall is identical, so one district's shortage report has to be an error
- (C) is fixed entirely by population, being larger only in the district with fewer people
- (D) is spread perfectly evenly across the Earth at all times regardless of local action

**15.** A factsheet pairs two numbers: India holds about 17% of the world's people but only roughly 4% of its fresh-water resources. Reasoning carefully from both figures together, the average Indian must draw from a fresh-water share that is:

- (A) much smaller than the world average per person — a large slice of the population sharing a far smaller slice of the water signals genuine water stress
- (B) far larger than the world average per person, signalling a comfortable surplus
- (C) exactly equal to the world average, so the two figures cancel and create no pressure
- (D) automatically expanding as the population grows, since more people attract more water

**16.** A borehole log reads: '0 to 7 m the pores held only air; from 7 m downward every pore was completely full of water; the porous water-bearing layer continued to 40 m.' Match the names correctly: what is the 7 m level, and what is the 7-to-40 m layer?

- (A) the 7 m level is the aquifer and the 7-to-40 m layer is the water table
- (B) the 7 m level is the water table (where pores first become fully water-filled), and the 7-to-40 m porous water-bearing layer is the aquifer
- (C) the 7 m level is the infiltration depth and the 7-to-40 m layer is a spring line
- (D) both the 7 m level and the whole 7-to-40 m layer are together called the water table

**17.** Four wells are bored on one site into four layers: thick porous sandstone that gushes water; solid unbroken granite that stays dry; dense clay that yields only a slow trickle; and a cracked limestone that both holds and freely passes water. Which two layers act most fully as aquifers?

- (A) the solid granite and the dense clay, because the hardest and the tightest rocks hold the most water
- (B) the porous sandstone and the cracked limestone, because each both stores groundwater and lets it move freely to a well
- (C) only the dense clay, because it traps water and refuses to let any of it move
- (D) all four equally, since any underground rock layer counts as an aquifer

**18.** Gentle rain falls for many hours onto open, vegetated soil with almost nothing running off, and a few days later the area's wells rise. A pupil credits the rise to 'condensation onto the cool soil grains.' The single process that actually carried the rain down to the groundwater is:

- (A) infiltration — the slow seeping of rain water downward through the pore spaces of the soil until it reaches and recharges the groundwater
- (B) condensation of vapour onto the cool soil grains, exactly as the pupil suggested
- (C) evaporation rising back up from the wet soil surface into the air
- (D) transpiration carrying the water up and out through the leaves of nearby plants

**19.** A hillside spring flows steadily all year, while a hand-dug well only a little downslope runs dry every summer. A bystander says, 'The well dries because it is deeper than the spring.' The most consistent chapter-based explanation is instead that:

- (A) the spring's outlet stays below the groundwater all year so it always flows, whereas the well's bottom sits above the summer water table, so it dries when the table drops
- (B) the well must be deeper than the spring, which is exactly why it empties first each summer
- (C) the spring manufactures its own water, so it never depends on the surrounding water table at all
- (D) summer rain somehow reaches the spring while skipping the nearby well entirely

**20.** On a gently rolling plain, neighbours find that under the low marshy hollows water is struck at shallow depth, while atop the higher rises wells must go far deeper before hitting water. The pattern most directly shows that the water table:

- (A) roughly follows the shape of the land surface, lying nearer the surface beneath low hollows and deeper beneath high ground
- (B) is perfectly flat everywhere underground, no matter how the surface above it rises and dips
- (C) sits deepest of all directly beneath the lowest hollows and shallowest beneath the highest rises
- (D) rises only under the tallest rises and is entirely absent beneath the low hollows

**21.** A farm replaces its shallow open dug well with a powerful deep tube well that reaches water no dug well could touch. The owner boasts, 'A deeper well raises the area's water table.' Considering the effect of the switch by itself, the truth is that going deeper:

- (A) raises the water table, because a deeper well pushes the stored water upward through the area
- (B) does not raise the table at all — it merely reaches lower water, and if it lets the farm pump still harder it can actually pull the table further down
- (C) instantly refills the whole aquifer simply by reaching previously untouched water
- (D) turns previously salty groundwater fresh as a side benefit of the greater depth

**22.** A town's aquifer gains 60 units of recharge and loses 60 units to pumping each year, holding steady. The town then triples its pumping while recharge stays at 60. Over the following years the stored water and the table will:

- (A) stay level, because tripling the pumping automatically triples the recharge to match
- (B) rise, because faster, harder pumping pulls extra rain down into the ground
- (C) fall just once in the first year and then settle back on its own to the old level
- (D) fall, and fall ever faster, because each year now removes 180 units while only 60 return, opening a widening 120-unit yearly deficit

**23.** Over four consecutive years a monitored aquifer's stored volume is logged as 100, 91, 79, 64 units, while the yearly recharge stayed fixed. The drops are 9, then 12, then 15 units. The most consistent reading is that withdrawal each year was:

- (A) less than recharge, so the stored volume should have been climbing, not falling
- (B) exactly equal to recharge each year, which would have kept the volume perfectly level
- (C) greater than recharge, and the yearly gap between them kept widening, so the store fell by a larger amount every year
- (D) zero throughout, since the volume can only fall when nothing is pumped out

**24.** Town P recharges 80 and withdraws 110 units a year; Town Q recharges 110 and withdraws 80. They share the same starting table and the same rainfall. After several years, the most accurate prediction is:

- (A) P's table falls because it runs a 30-unit yearly deficit, while Q's rises because it runs a 30-unit yearly surplus
- (B) both tables stay identical, because the two towns' numbers are mirror images of each other
- (C) Q's table falls faster, because its larger recharge stirs up and disturbs the aquifer more
- (D) P's table rises, because pumping a larger amount each year brings more water to the surface

**25.** A region's annual recharge has held perfectly steady for years, but each season a few more identical bore wells are sunk, every one pumping the same amount as the rest. Year on year, the gap between water withdrawn and water recharged:

- (A) narrows, so the water table slowly recovers toward its original level
- (B) stays constant, leaving the water table unchanged from one year to the next
- (C) widens, because withdrawal climbs each year while recharge stays fixed, pushing the water table progressively lower
- (D) vanishes, because adding more wells forces a matching increase in recharge

**26.** City G keeps its annual pumping rate exactly unchanged for a decade, yet its water table, once stable, now falls year after year. Over the same decade the city paved most of its open ground for malls and roads. The most coherent explanation is that:

- (A) the paving cut recharge, so the same unchanged pumping now outpaces a reduced inflow, opening a deficit that lowers the table
- (B) the city must secretly be pumping far more than its records admit, despite the steady official rate
- (C) the paving made the groundwater salty, and salty water naturally sinks to a lower level
- (D) the region's rainfall must have collapsed at the very same time as the paving

**27.** A village's open dug well stayed full for thirty years but ran completely dry this summer, even though last year's monsoon was perfectly normal. Across the same years the surrounding area added scores of tube wells. The most likely chapter-based cause is that:

- (A) one ordinary monsoon last year somehow permanently destroyed the aquifer beneath the well
- (B) years of over-extraction across the area have gradually lowered the regional water table until it dropped below this well's bottom
- (C) the well's water evaporated straight upward through the surrounding solid rock
- (D) the well abruptly turned salty, and salty water cannot remain inside a dug well

**28.** A district's leaders point to three worsening problems — shallow wells drying up, needing ever-deeper tube wells, and steeply rising pumping bills — and declare these to be the CAUSE of their water crisis. Their reasoning error is that these three are actually:

- (A) symptoms of an already-falling water table, not its cause; the table is dropping first, and these hardships follow from it
- (B) the genuine causes, so simply drilling the wells deeper will solve the whole crisis
- (C) completely unrelated to the water table and only a coincidence occurring at the same time
- (D) clear signs that the water table is in fact rising far too quickly

**29.** Coastal wells that gave sweet fresh water for decades slowly turn brackish after years of intensive pumping along the shore. Within the chapter's logic, the salinity arises because:

- (A) the act of pumping chemically transforms the fresh water in the well into salt water
- (B) the coastal rainfall is naturally salty and has been refilling the wells with brine
- (C) the concrete buildings near the coast manufacture salt within the surrounding soil
- (D) drawing down the fresh groundwater lowered its level enough that nearby salty water moved inland toward the over-pumped wells

**30.** A water-stressed town's audit concludes that every one of its problems traces back to a single root: it draws more groundwater each year than the rains can replace. Which single response actually treats that CAUSE rather than merely its symptoms?

- (A) fitting every well with a more powerful pump to keep lifting the ever-deeper water
- (B) cutting back the over-extraction while boosting recharge, so that withdrawals no longer outrun the inflow



- (C) repeatedly deepening each well to chase the dropping water further down
- (D) trucking in bottled water to cover the town's daily drinking needs

**31.** Of these four scenarios, which would drive a region's water table DOWN the fastest over just a few years?

- (A) plentiful rainfall onto open soil that is dotted with many new recharge pits
- (B) scanty rainfall combined with a booming population, new water-hungry factories, and widespread concreting of open ground
- (C) every farm switched to drip irrigation while every rooftop harvests its rain
- (D) a programme of tree planting, leak repairs, and keeping all open ground unpaved

**32.** Two identical adjoining plots receive the same rain for several monsoons. Plot M is left as open garden soil; Plot N is fully tiled, with its rain piped straight to a street storm drain. A pupil argues, 'Plot N's tiles trap the rain underground.' After the monsoons the water table is in fact higher under:

- (A) Plot N, because the tiles trap and store the rain underground exactly as the pupil claims
- (B) Plot N, because the storm drain carries the collected water back down into the aquifer
- (C) neither, because tiled ground and open soil recharge the groundwater at the same rate
- (D) Plot M, because open soil lets the rain infiltrate and recharge the groundwater, whereas Plot N's rain is carried away before it can soak in

**33.** A planner insists, 'Paving every open lot cannot hurt our groundwater, because exactly the same amount of rain still falls on the city as before.' The decisive flaw in this argument is that paving:

- (A) actually increases how much of the rain soaks down through the sealed surface
- (B) turns the falling rain salty before it can ever reach and recharge the soil
- (C) has no effect on recharge at all, as long as the total rainfall is unchanged
- (D) blocks the rain from soaking in, so the same rainfall now runs off into drains and no longer recharges the aquifer

**34.** After a hillside forest is cleared, valley wells below begin to struggle within a few seasons, though rainfall is unchanged. The pupil who reasons best lays out the causal chain as:

- (A) fewer trees hold back and slow the rain less, so more runs off and less soaks in, recharge to the valley aquifer drops, and the valley water table falls
- (B) fewer trees cool the soil, so the rain freezes in the ground and the wells ice over
- (C) fewer trees stop the rain falling altogether, so no water at all reaches the valley
- (D) fewer trees turn the falling rain salty, so the valley wells become undrinkable

**35.** An orchard that for years kept a valley well brimming is bulldozed for a tarmac car park, and within a few summers that well struggles. Beyond losing the car park's runoff control, why does the loss of the trees themselves matter to the well?

- (A) the trees had been pumping groundwater upward to keep the well topped up directly
- (B) the trees had been creating brand-new water inside their roots and feeding it to the aquifer
- (C) the trees had been blocking the rain from ever reaching the soil, so removing them should help the well
- (D) the trees had been helping rain slow down and soak in, supporting the recharge that kept the valley well fed

**36.** A household pipes its rooftop rain down into a gravel-and-sand recharge pit so the water seeps slowly into the ground, instead of letting the same rain pour into a sealed street drain. This pit raises the water table more than the drain would because it:

- (A) chemically transforms the rooftop rain into fresh groundwater inside the gravel
- (B) pumps stored water upward out of the soil so the household can use it later
- (C) lets the collected rain infiltrate down into the aquifer, whereas the drain would carry that same water away as runoff
- (D) heats the rain so the warmer water naturally sinks deeper into the ground on its own

**37.** Across dry parts of India, traditional structures such as the bawri (step-well), the tanka, and the johad were built chiefly to:

- (A) collect, store, or recharge rain and groundwater so it could be used through the dry season
- (B) generate electricity from the force of falling water
- (C) remove the salt from sea water for use by inland villages
- (D) measure and record the exact daily rainfall of each region

**38.** A village throws an earthen johad across a seasonal stream. The chief way this lifts the local water table is by:

- (A) pumping the stored groundwater up to the surface for the village to draw on
- (B) sealing the streambed completely so that no water can ever soak through it
- (C) spreading the collected water into a wide thin sheet so it evaporates quickly
- (D) holding back the flowing rain water so it has time to seep into the ground and recharge the aquifer

**39.** A farmer converts a field from flood irrigation to drip irrigation on the same soil with the same crop. Compared with before, the field now:

- (A) uses much more water than flooding and so lowers the water table even faster
- (B) needs no irrigation water at all once the drip pipes have been laid out

(C) uses far less water by delivering it drop by drop straight to the roots, easing the demand placed on the groundwater

(D) causes the surrounding groundwater to turn salty purely as a side effect of dripping

**40.** Besides cutting total water use, drip irrigation usually leaves far fewer weeds in the dry strips between crop rows. The reason within the chapter's logic is that:

(A) the drip water carries a built-in chemical that kills the weeds it touches

(B) the slow drops flood the rows and physically wash all the weeds away

(C) only each crop plant's root zone is kept moist, so weeds in the dry gaps between rows receive little water and struggle to grow

(D) drip irrigation chills the bare soil so cold that no weed seed can sprout

**41.** A nursery worker must water 300 seedlings in individual pots on a scorching afternoon using as little water as humanly possible. Of these methods, the one that best meets the goal is:

(A) flooding the whole bench until water reaches every pot from below

(B) an overhead sprinkler left running across all the pots the entire afternoon

(C) hosing water generously over the leaves of every seedling in turn

(D) a drip line releasing water slowly right at the base of each pot, wetting only the roots

**42.** Rooftop rainwater harvesting is pushed especially hard in big cities. The strongest chapter-based reason is that:

(A) cities by their very nature receive far more rainfall than any village does

(B) city roofs and pavements shed enormous volumes of rain that would otherwise run straight off into drains without ever recharging the ground

(C) city groundwater is effectively endless, so harvesting is only a harmless precaution

(D) harvesting works at all only in places that have no soil whatsoever

**43.** A city planner must pick the single most reliable structural measure to slow a steadily falling water table. The best choice is:

(A) sanctioning many more bore wells so that the heavy demand is shared out among them

(B) tiling over the city's remaining open grounds to keep them tidy and clean

(C) requiring rainwater-harvesting recharge structures on new buildings so rain soaks back into the ground

(D) routing all the city's storm water out and away as quickly as possible

**44.** A town's water-use survey reads: drinking and cooking 5%, washing and bathing 22%, irrigation of nearby farms 65%, industry 8%. To ease groundwater stress the most, the town should first target:

- (A) drinking and cooking, since trimming the very smallest use somehow helps the most
- (B) industry, treating its 8% as if it were the largest share to cut
- (C) irrigation, since at 65% it is by far the largest use and can shift to water-saving drip methods
- (D) nothing, on the assumption that none of these uses can possibly be reduced

**45.** Concreting a floodplain that used to soak up the monsoon typically produces which twin effect during and after the rains?

- (A) less flooding during the rain and more recharge into the ground afterwards
- (B) no flooding at all during the rain and no change whatever in the recharge
- (C) more surface flooding while the rain falls, and less groundwater recharge afterwards
- (D) more recharge into the ground during the rain but also far more flooding

**46.** A region installs rooftop harvesting on all buildings (raising recharge) and converts every farm to drip (cutting withdrawal), so that recharge now clearly exceeds withdrawal. A skeptic warns, 'Two big interventions will only drain the aquifer faster.' Over time the water table will actually:

- (A) fall, because two interventions working together drain the aquifer faster, as the skeptic warns
- (B) rise, because more water now enters the aquifer than leaves it, the very definition of a surplus
- (C) stay perfectly flat, since recharge and withdrawal are forced to always match exactly
- (D) rise only for as long as the two measures are later removed again

**47.** Farm A floods its fields and its well runs dry by April; neighbouring Farm B drips its fields, on the same soil with the same rainfall, and its well lasts until June. The fairest conclusion from the comparison is that:

- (A) Farm B's lower water use, thanks to drip, let its well outlast Farm A's flooded fields
- (B) flood irrigation conserves more groundwater than drip irrigation does
- (C) the choice of irrigation method has no bearing on how long a well lasts
- (D) Farm B must simply have received much more rain than Farm A that year

**48.** A school finds dozens of taps each leaking only slightly, together losing about 30 litres a day. A teacher shrugs, 'One little drip is far too small to bother with.' The most sensible response, in the chapter's spirit, is to:

- (A) ignore them, since a single small drip really is far too little to matter at all
- (B) open every leaking tap fully so that it can no longer waste water by dripping
- (C) leave a bucket under each leaking tap permanently to catch the drips forever
- (D) mend the leaks promptly, because many small drips add up to a large daily loss of precious fresh water

**49.** Which set of household actions are ALL genuine fresh-water conservation measures?

- (A) taking long showers, leaving the tap running while soaping, and filling the bathtub to the brim
- (B) paving over the garden, flood-irrigating the lawn, and leaving the taps dripping
- (C) draining the rooftop rain to the road, over-watering the pots, and running the hose freely
- (D) bathing with a bucket and mug, closing the tap while brushing, and promptly mending leaking taps

**50.** A town's reservoir reads full, then a little down, then lower, then lowest across four years. Rainfall stayed normal each year, but the town drilled many new tube wells annually. The pattern most strongly indicates that:

- (A) recharge has outpaced withdrawal, so the stored water should have been climbing instead
- (B) rainfall must in fact have failed completely in every one of those four years
- (C) withdrawals have steadily outpaced recharge, progressively depleting the stored water despite normal rainfall
- (D) the falling level proves the stored water has been quietly turning into vapour

**51.** Which single observation would be the clearest sign that an area's groundwater recharge programme is genuinely WORKING over a few seasons?

- (A) the nearby ocean gradually becomes noticeably less salty year by year
- (B) the monsoon rainfall suddenly doubles compared with previous years
- (C) wells that had gone dry begin yielding water again, and at shallower depths than before, over several seasons
- (D) the local rivers stop flowing entirely during every dry season

**52.** A district official boasts, 'Double our tube wells and we permanently double our water supply.' Drawing on the chapter, the fundamental flaw is that:

- (A) tube wells only withdraw from a limited aquifer, so pumping faster than it recharges soon lowers the table and dries the wells
- (B) adding more tube wells actually creates brand-new water out of nothing
- (C) the aquifer is effectively unlimited, so the doubling plan is perfectly safe forever
- (D) tube wells add fresh rain to the sky above, keeping themselves topped up indefinitely

**53.** A village runs four measures together for several years: rooftop harvesting, drip irrigation on its farms, repairing leaking mains, and planting trees on open ground. Over time its water table most likely:

- (A) falls faster than before, because building more structures must mean using more water overall
- (B) turns salty as an unavoidable combined side effect of running the four measures at once

(C) rises, because recharge improves through harvesting and trees while wasteful withdrawals fall through drip and leak repair

(D) stays exactly the same forever, completely untouched by any of the four measures

**54.** Two villages receive identical rainfall yet, after a decade, end up with markedly different water tables. The best explanation faithful to the chapter is that the villages may differ in:

(A) nothing at all — equal rainfall must force their water tables to end up identical

(B) which village has more concrete, since concrete always raises the water table

(C) how much groundwater they pump out, and how much of their rain they let infiltrate versus allow to run off

(D) how deeply they dig their wells, since digging deeper wells raises the table

**55.** During a prolonged water shortage, a row of vegetable plants is left without enough water and the season's yield falls sharply. The chapter's explanation is simply that:

(A) plants need water to grow and produce, so a shortage stunts their growth and lowers the crop yield

(B) less water makes the plants grow faster and yield even more than usual

(C) the plants convert the missing water into extra vapour and so thrive without it

(D) a water shortage makes the plants saltier rather than making the harvest smaller

**56.** Two aquifers start at the same level. Aquifer X loses 10 units of storage in year one, 10 in year two, and 10 in year three (a steady deficit). Aquifer Y loses 5, then 10, then 20 (an accelerating deficit). After three years, comparing them tells us that:

(A) X has fallen further overall, because a steady loss adds up to more than an accelerating one

(B) both have fallen by exactly the same total, so their water tables are now identical

(C) Y has actually risen, because an increasing yearly figure means water is being added

(D) Y has fallen further in total and is dropping faster each year, so its withdrawal-over-recharge gap is widening while X's gap stays constant

**57.** A pupil claims that because water vapour keeps rising into the sky, 'Earth is steadily and permanently losing its water to the air.' The chapter-accurate correction is that:

(A) the vapour does escape upward forever, so Earth is indeed slowly drying out over the ages

(B) the rising vapour is destroyed high in the air, which is why the total water keeps shrinking

(C) the vapour becomes a brand-new substance aloft and never returns to Earth as water

(D) the vapour does not leave for good — it later condenses and falls back as precipitation, so water only changes form and place while the planet's total stays roughly constant

**58.** A salesman markets a 'water-creating' rooftop unit, claiming it 'manufactures new fresh water from the rain that hits your roof, adding to Earth's total supply.' Judged against the chapter, the honest description of what such a unit can really do is:

- (A) it genuinely manufactures brand-new water, adding to the fixed global total
- (B) it converts the salty sea breeze into fresh drinking water on the rooftop
- (C) it can only collect and redirect rain that already exists — capturing roof runoff for use or recharge — without creating any new water or changing Earth's total
- (D) it destroys excess rain so that less water has to fall on the city below

**59.** World Water Day is observed each year to remind everyone to value and conserve water. It falls on:

- |                 |              |
|-----------------|--------------|
| (A) 5 June      | (B) 22 April |
| (C) 14 November | (D) 22 March |

**60.** Which statement best captures the chapter's central message that 'water is a precious resource'?

- (A) water is effectively unlimited, so making any effort to save it is pointless
- (B) usable fresh water is genuinely scarce, so we must use it wisely and manage it carefully
- (C) only farmers, not ordinary households, ever need to bother conserving water
- (D) the sea will always supply more than enough drinking water for everyone

# Solutions — Science (Class 7), Chapter 16: Water: A Precious Resource — Paper 5

## Paper 5 — (progressively harder)

Marking: +4 correct, –1 wrong, 0 unattempted. Maximum marks **240**.

## Answer Key

| Q  | Ans | Q  | Ans | Q  | Ans | Q  | Ans | Q  | Ans | Q  | Ans |
|----|-----|----|-----|----|-----|----|-----|----|-----|----|-----|
| 1  | D   | 11 | A   | 21 | B   | 31 | B   | 41 | D   | 51 | C   |
| 2  | B   | 12 | C   | 22 | D   | 32 | D   | 42 | B   | 52 | A   |
| 3  | B   | 13 | B   | 23 | C   | 33 | D   | 43 | C   | 53 | C   |
| 4  | D   | 14 | A   | 24 | A   | 34 | A   | 44 | C   | 54 | C   |
| 5  | B   | 15 | A   | 25 | C   | 35 | D   | 45 | C   | 55 | A   |
| 6  | B   | 16 | B   | 26 | A   | 36 | C   | 46 | B   | 56 | D   |
| 7  | B   | 17 | B   | 27 | B   | 37 | A   | 47 | A   | 57 | D   |
| 8  | D   | 18 | A   | 28 | A   | 38 | D   | 48 | D   | 58 | C   |
| 9  | A   | 19 | A   | 29 | D   | 39 | C   | 49 | D   | 59 | D   |
| 10 | C   | 20 | A   | 30 | B   | 40 | C   | 50 | C   | 60 | B   |

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

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## Worked Solutions

**1. (D)** Track the states: ocean liquid to atmospheric vapour is one change; vapour freezing into glacier snow is a second; glacier ice melting into river liquid is a third. The river-to-sea step is liquid staying liquid, so it adds no state change, giving three in total. Crucially, every step only rearranged the same molecule into a new form and place; nothing was manufactured by the snowfall, nothing was annihilated, and there is no need to invoke loss to space. That conservation across rearrangement is precisely the chapter's reason the planet's water total barely changes year to year.

**2. (B)** A sealed terrarium is a tiny model of the global cycle: water leaves the soil as vapour (evaporation), the vapour meets the cool glass and condenses to mist, and that liquid drips back to the soil — a precipitation-like return. Because the jar is sealed, the same water simply goes round and round with no net gain or loss, mirroring why Earth's total stays constant. Plants release water they first drew up, they do not manufacture it; sealed glass loses nothing; and the glass droplets are the very same water that evaporated, not a new sunlight-made substance.



**3. (B)** Begin with the frost present at dawn: it is solid ice. By midday it has become liquid beads, so the first change is melting (solid to liquid). By dusk the rail is dry, meaning the liquid has gone to vapour — evaporation (liquid to vapour) is second. Overnight new frost grows, so vapour has gone directly back to ice, the solid-forming return that closes the day. The order is therefore melt, evaporate, then re-form as ice. Putting evaporation or condensation first contradicts the fact that frost (ice) was the starting state, and freezing-first ignores that the frost already existed at dawn.

**4. (D)** The white plume you can see above a pot is not vapour at all — true water vapour is invisible. The plume is vapour that has already cooled enough to condense into a mist of tiny liquid droplets. The genuinely gaseous water is the invisible dampness near the spout, which only reveals itself when it later condenses on a cold tile. So both are one substance, water, in different states, and the child has it backwards: the invisible part is the gas, the visible plume is liquid. The dampness is not air, and neither is a separate chemical.

**5. (B)** The skater rests on solid ice, fish move through liquid water beneath, and breath-fog is condensed vapour above — three states of the identical substance coexisting because temperature and exposure differ from the surface down into the water and up into the air. This shows water can occupy all three states simultaneously when local conditions differ. It is not forced into one state per body of water, the three forms are all just water rather than separate chemicals, and ice plainly forms far from the poles, here directly over liquid water.

**6. (B)** In the open sky, air rises, expands and cools, and that cooling is what condenses vapour into cloud droplets. But rising is only the means of cooling, not the cause of condensation itself — the cause is the vapour reaching a low enough temperature. The flask achieves the same cooling directly by chilling the walls, so droplets form without any rising air, which is fully consistent with the cloud rule rather than contradicting it. Heating would not condense vapour, and ordinary chilled-wall droplets are liquid condensation, not ice.

**7. (B)** The orchard's advantage is transpiration: many trees pull water up from the soil and release it as vapour through their leaves, so a living canopy adds far more moisture to the air than bare gravel. Condensation would remove vapour from the air (turning it to liquid), not add it, so the pupil's reasoning is the wrong direction. Infiltration moves water downward into the soil, not upward into the air, and orchards do not conjure their own private rainfall. Only transpiration explains the moisture surplus.

**8. (D)** A loop starting at the sea must begin with evaporation: the parcel rises off the sea as vapour. Aloft it cools and condenses into a cloud. The cloud then releases the water as rain that lands on the hill. Finally the rain on the ground seeps downward through the soil toward the groundwater (infiltration). The only self-consistent order is rise, condense, rain, seep. Every other option places condensation or seeping before evaporation, which is

impossible because the parcel cannot condense or infiltrate before it has first left the sea as vapour.

**9. (A)** Sleet, drizzle, hail and snow all share one thing: each delivers water from the atmosphere down to the ground, which is the definition of precipitation. Their differing states (solid hail and snow, liquid drizzle) and speeds are irrelevant to the classification. They are not evaporation, which moves water upward off the ground; they are not condensation, which is the cloud-forming step that precedes the fall, not the fall itself; and they are not transpiration, which is moisture released by plant leaves.

**10. (C)** The 71% figure is a surface-area statistic — how much of the planet's face is covered by water — and says nothing about quality. The 97% figure tells the real story of the total volume: almost all of it is salty ocean, leaving only a small fresh fraction, most of which is frozen or deep underground. The student's error is treating surface coverage as if it equalled usable supply. Surface coverage does not imply freshness, oceans are salty not drinkable, and the drinkable fraction is far below 71% — it is tiny.

**11. (A)** Scaling the real proportions to 1000 jugs: roughly 970 are salty ocean and only about 30 are fresh. Within those 30, the bulk is frozen in ice-caps and glaciers, so the readily usable groundwater and surface water is a small slice of an already small fraction. The 710 answer wrongly imports the surface-coverage percentage as if it described volume and freshness. A 500-500 split grossly overstates fresh water. And '970 fresh, 30 salty' simply inverts the true figures, since the oceans hold the overwhelming majority.

**12. (C)** The guide's first claim is correct: most of Earth's fresh water really is stored in ice-caps and glaciers. But the conclusion does not follow, because that ice is frozen and largely inaccessible, so it cannot be the everyday tap source. Day-to-day usable water is drawn mostly from groundwater and surface sources. Hence one half is true and one false. The options that call both halves true, both false, or reverse which half is true all misjudge either the storage fact or the accessibility fact.

**13. (B)** The oceans dominate every other store by far, so they top the list. Among the fresh stores, the chapter teaches that most fresh water is frozen in ice-caps and glaciers, placing them next, ahead of groundwater. River water flowing at any instant is a very small store and comes last. The correct descending order is oceans, ice/glaciers, groundwater, rivers. Putting rivers first inverts the scale entirely; the other wrong orders either rank groundwater above the ice or place the oceans below the ice, both of which contradict the chapter's proportions.

**14. (A)** Equal rainfall does not guarantee equal usable water, because availability also turns on how much water each district stores, how heavily it pumps, and how much rain infiltrates to recharge rather than runs off. District B may over-extract or pave its ground, draining its table despite the same rain. So the lesson is uneven distribution shaped by local management, not rainfall alone. The claim that equal rain forces equal water ignores

withdrawal and recharge; population alone does not fix supply; and water is demonstrably not spread evenly everywhere.

**15. (A)** Compare shares: 17% of people but only 4% of the fresh water means each person's slice is well below the global average — many people dividing a much smaller resource. That mismatch is the textbook signature of water stress. A larger-than-average share would require more water per person, the opposite of these numbers. The shares are not equal, so they do not cancel. And population growth does not summon more water; it only divides the fixed share among still more people, worsening the stress.

**16. (B)** The water table is the upper level at which the pore spaces first become completely filled with water — exactly the 7 m mark here. The aquifer is the porous, water-bearing layer of rock or gravel that stores and yields that groundwater — the 7-to-40 m zone. So the level is the table and the layer is the aquifer. Swapping the two names confuses a single surface with a thick layer; 'infiltration depth' and 'spring line' are not the chapter's terms for these; and the water table is a level, not the whole layer.

**17. (B)** An aquifer must do two things: store water and let it move freely to a well, which requires connected pore space or cracks. The porous sandstone and the fractured limestone both qualify — they hold water and yield it readily. Solid granite is non-porous and stays dry, so it is not an aquifer; dense clay may hold water but barely lets it move, so it yields only a trickle and acts more as a barrier. Not every layer is an aquifer — only those that both store and transmit water do.

**18. (A)** Rain reaching the groundwater does so by infiltration: it seeps slowly downward through the soil pores until it joins the saturated zone and recharges the wells, which is why they rose. Condensation would form liquid from vapour on a surface, not move rain downward, so the pupil's idea is wrong. Evaporation moves water upward off the soil into the air, the reverse of recharge. Transpiration sends water out through leaves into the atmosphere, also upward and away, not down to the groundwater.

**19. (A)** Whether a well or spring holds water depends on its position relative to the water table. The spring discharges below the table all year, so it keeps flowing; the well's base lies above the lowered summer table, so it goes dry until the table rises again. Depth alone does not decide this — a deeper well would actually reach water longer, contradicting the bystander. Springs do not create water independently of the table, and the same rain that recharges one feeds the whole local table, not one spot selectively.

**20. (A)** The water table is not a dead-flat plane; it broadly mimics the land above, sitting closer to the surface under low spots and dropping deeper under high ground — which is exactly why hollows reach water shallowly and rises require deep wells. A perfectly flat table would make the depths equal, contradicting the observation. The pattern is the reverse of 'deepest under hollows', and the table is certainly not absent under the low areas — that is where it is shallowest.

**21. (B)** Depth changes where a well taps water, not how much water the aquifer holds. Reaching deeper does nothing to raise the table; if anything, enabling heavier pumping draws the table down faster. A deeper well does not push water upward across the area, it does not create or refill stored water, and depth has no power to desalinate groundwater. So by itself the switch reaches lower water and may worsen, never raise, the table.

**22. (D)** Balance is recharge minus withdrawal. At 60 in and 60 out the table held. Tripling withdrawal to 180 against an unchanged 60 recharge creates a 120-unit annual deficit, so the store loses 120 each year and the table falls steadily, the loss compounding as the aquifer empties. Pumping does not summon matching recharge — recharge depends on rain and infiltration, not on how hard you pump. Harder pumping does not pull extra rain down, and a system in permanent deficit does not self-correct back to the old level.

**23. (C)** Each year the store drops, so withdrawal must exceed the fixed recharge — outflow beats inflow. The losses grow (9, 12, 15), meaning the deficit itself is widening year on year, perhaps from more wells or harder pumping. If withdrawal were below recharge the store would rise; if it equalled recharge the store would stay flat; and zero withdrawal would let recharge build the store up, not draw it down. Only a widening withdrawal-over-recharge deficit produces an accelerating fall.

**24. (A)** Net change is recharge minus withdrawal. Town P: 80 minus 110 is a 30-unit deficit each year, so its table falls. Town Q: 110 minus 80 is a 30-unit surplus, so its table rises. The numbers being mirror images means the two tables move in opposite directions, not that they stay identical. Larger recharge fills, it does not 'disturb' and lower, a table, so Q does not fall. And pumping more does not raise a table — withdrawal lowers it, which is why P, the heavier pumper, declines.

**25. (C)** Recharge is fixed, but each new well adds to the total withdrawal, so the outflow rises year after year while the inflow does not. The withdrawal-minus-recharge gap therefore widens annually, driving the table steadily lower. It cannot narrow while withdrawal only grows; it cannot stay constant when wells keep being added; and adding wells does nothing to increase recharge, which depends on rainfall and infiltration, not on how many wells draw from the ground.

**26. (A)** If withdrawal is genuinely unchanged, a newly falling table must mean inflow has dropped. Paving over open ground blocks rain from infiltrating, slashing recharge, so the same steady pumping now exceeds the smaller inflow and the table declines — a textbook cause. There is no need to assume hidden over-pumping when reduced recharge fully explains it. Salinity does not lower a table by 'sinking', and inventing a simultaneous rainfall collapse is an unsupported extra assumption when paving alone accounts for the deficit.

**27. (B)** A long-full well going dry after years of nearby tube-well expansion points to chronic over-extraction lowering the whole area's water table until it sank beneath the

well's base, so the well now reaches only dry ground. A single normal monsoon does not wreck an aquifer; if anything it recharges it. Water does not evaporate up through solid rock to empty a well, and turning salty would change the water's quality, not cause the well to hold no water at all.

**28. (A)** Drying shallow wells, deeper drilling and costlier pumping all flow from one underlying fact: the water table is falling. They are downstream symptoms, not the root cause, which is over-extraction outrunning recharge. Treating symptoms as causes leads to chasing the water deeper rather than fixing the imbalance. They are not the true cause, so deeper drilling only postpones the problem; they are tightly linked to the table, not coincidental; and they signal a falling, not a rising, table.

**29. (D)** Near a coast, fresh groundwater normally sits above or beside salty water. Heavy pumping lowers the fresh supply so much that the adjacent salty water encroaches toward the wells, turning them brackish. The cause is the lowered fresh table letting saline water move in. Pumping is a physical withdrawal, not a chemical reaction that creates salt; rain is fresh, not naturally briny; and concrete does not manufacture salt in soil. Only the inward movement of nearby salt water fits the chapter's reasoning.

**30. (B)** The root cause is withdrawal exceeding recharge, so the only true fix is to close that gap — pump less and recharge more — restoring balance to the table. Stronger pumps, deeper wells and imported bottled water all leave the imbalance untouched: the first two simply chase the falling water (a symptom) and accelerate depletion, while trucking water sidesteps the aquifer problem entirely without healing it. Only reducing extraction and raising recharge addresses the cause.

**31. (B)** The fastest fall needs minimal inflow and maximal outflow at once: scanty rain starves recharge, a booming population and new factories raise withdrawal, and concreting blocks what little rain falls from soaking in — every factor pushing the table down together. The other three all favour the table: heavy rain on pitted open soil boosts recharge; drip plus harvesting cuts withdrawal and lifts recharge; and trees, leak repair and open ground all help recharge and reduce waste, raising rather than lowering the table.

**32. (D)** Recharge depends on rain soaking into the ground. Plot M's open soil lets water infiltrate to the aquifer, so its table rises. Plot N's tiles seal the surface and its rain is piped to a storm drain that whisks it away, so almost none recharges — its table stays lower. Tiles block, not trap, infiltration; a storm drain removes water from the site rather than returning it to the aquifer; and sealed and open ground plainly do not recharge equally, which is the whole point.

**33. (D)** What matters for groundwater is not how much rain falls but how much infiltrates. Paving seals the surface, so rain that once soaked in now sheets off into drains and leaves the area, cutting recharge even though the rainfall total is unchanged — that is the planner's blind spot. Paving reduces, not increases, infiltration; it does not chemically

salt the rain; and it certainly does affect recharge, since sealing the ground is precisely what stops the soaking-in that feeds the aquifer.

**34. (A)** Forests slow runoff and let rain soak in. Remove the trees and rain rushes off the bare slope instead of infiltrating, so less water recharges the valley aquifer and its table falls, starving the wells — all consistent with unchanged rainfall. Trees do not control whether soil water freezes, and clearing them does not make wells ice over. Deforestation does not stop rain from falling, only from soaking in. And it does not make rain salty; the issue is reduced infiltration, not contaminated water.

**35. (D)** Living trees and their litter slow runoff and encourage rain to infiltrate, which recharges the groundwater that feeds the valley well; removing them lets rain run off the tarmac instead, so recharge falls and the well struggles. Trees do not pump groundwater up to refill a well, nor do they manufacture new water in their roots — water is conserved, not created. And trees do not block rain from the soil; they help it soak in, so losing them hurts rather than helps the well.

**36. (C)** A recharge pit works by giving rooftop rain a place to seep slowly into the soil and reach the aquifer, adding to the groundwater — exactly the recharge a sealed drain prevents by sweeping the water off-site. The pit does not chemically convert rain; rain is already water, it just needs to infiltrate. It does not pump water upward; it sends water downward. And it does not rely on heating water to sink — infiltration through gravel and sand, not temperature, drives the recharge.

**37. (A)** These traditional structures share a single purpose: capturing and conserving water — storing rain and recharging groundwater — so communities had water during dry months. They were not built to generate hydroelectric power, which is unrelated to their design and scale; they were not desalination plants, since inland villages dealt with rain and groundwater, not sea water; and they were storage and recharge works, not rain gauges for measuring rainfall.

**38. (D)** A johad is a check-dam: by detaining the stream's flow it gives the water time to infiltrate the streambed and surrounding soil, recharging the aquifer and raising the table. It is not a pump and lifts no water to the surface; sealing the bed would do the opposite of recharge by preventing infiltration; and spreading water thin to evaporate it would lose the water to the air rather than send it underground. Detention for infiltration is the mechanism.

**39. (C)** Drip irrigation targets water precisely at each plant's root zone, so almost none is lost to runoff or evaporation from bare ground, and the field uses far less water than flooding — relieving pressure on the groundwater. It does not use more water than flooding; that is backwards, since flooding is the wasteful method. Crops still need some water, so 'none at all' is wrong. And dripping water onto crop roots does not salinate the surrounding groundwater.

**40. (C)** Drip delivers water only to the crop's roots, leaving the inter-row strips dry, so weeds there get little water and grow poorly — fewer weeds as a free bonus of targeted watering. The drip water is plain water, not a chemical weedkiller; drip releases small drops slowly and does not flood or wash anything away (that would be the wasteful method it replaces); and drip does not freeze the soil. The weed suppression is simply a consequence of keeping the gaps dry.

**41. (D)** Drip delivers water precisely to each pot's roots with minimal loss, so it wastes the least and best meets the save-water goal. Flooding the bench soaks the whole surface, wasting water on everything but the roots; an all-afternoon sprinkler loses huge amounts to evaporation and to gaps between pots; and hosing the leaves wets foliage and surroundings rather than roots, again wasting water. Only the drip line concentrates the scarce water exactly where the plants need it.

**42. (B)** Cities are covered in roofs and paving, so rain that lands on them cannot infiltrate and instead pours into drains and is lost; harvesting captures that runoff and routes it to recharge or storage, which is why it is promoted so strongly there. Cities do not inherently get more rain than villages; their groundwater is finite, not endless, which is exactly why harvesting matters; and harvesting is not limited to soil-less places — it is valuable precisely where sealed surfaces would otherwise waste the rain.

**43. (C)** To slow a falling table you must raise recharge or cut withdrawal; mandating harvesting recharge structures directly increases the rain returning to the aquifer, making it the reliable choice. More bore wells only add to withdrawal and pull the table down faster. Tiling open ground seals the very surfaces through which rain recharges, worsening the fall. And rushing storm water out of the city throws away the rain that could have soaked in. Only the recharge requirement helps.

**44. (C)** Targeting the biggest user yields the biggest savings: irrigation at 65% dwarfs every other category, and switching from flood to drip can cut that use sharply, easing the stress most. Drinking and cooking at 5% is the smallest slice, so squeezing it saves little. Industry at 8% is far from the largest, so calling it the biggest is simply misreading the table. And claiming nothing can be reduced ignores the well-known water savings drip irrigation offers on the dominant irrigation use.

**45. (C)** A natural floodplain absorbs monsoon water, both easing floods and recharging groundwater. Concreting it seals the surface, so during the rains water cannot soak in and instead pools and floods, while afterwards little has infiltrated, so recharge drops — a damaging twin effect. The reverse pairing (less flooding, more recharge) describes the unpaved floodplain. Sealing the surface cannot leave flooding and recharge unchanged, and it cannot increase recharge while also flooding, since blocked infiltration is the common cause of both.

**46. (B)** The two measures push the balance in the same favourable direction: harvesting raises recharge and drip lowers withdrawal, so inflow now exceeds outflow and the surplus lifts the table. The skeptic has it backwards — these measures add water and save water, they do not drain the aquifer. Recharge and withdrawal are not forced to match; here they are deliberately unbalanced toward surplus. And the rise comes from keeping the measures running, not from removing them, which would simply undo the surplus.

**47. (A)** With soil and rainfall held equal, the only difference is the irrigation method, so the longer-lasting well on Farm B must come from drip's lower water use, which drew down the shared groundwater more slowly. Flood irrigation is the wasteful method, so it conserves less, not more, than drip. The method clearly does affect well life here, contradicting the 'no bearing' option. And the rainfall was stipulated identical, so Farm B cannot have received more rain — the method is the deciding factor.

**48. (D)** Individually a drip seems trivial, but dozens of taps summing to 30 litres a day is a serious cumulative loss of scarce fresh water, so the right move is to repair the leaks promptly. Ignoring them lets the daily waste continue; opening taps fully wastes far more water, not less; and parking a bucket under each tap merely collects the loss without stopping it and still squanders the running water. Fixing the leaks is the only response that conserves.

**49. (D)** Each action in the correct set saves water: a bucket-and-mug bath uses far less than a shower, closing the tap while brushing stops clean water running away, and mending leaks ends a steady loss. The other sets are all the opposite — long showers, running taps and full bathtubs waste water; paving, flood-irrigation and dripping taps waste water and harm recharge; and draining roof rain away, over-watering and running hoses all squander it. Only the first set conserves throughout.

**50. (C)** Normal rainfall rules out a recharge failure, so the steady fall must come from rising withdrawal — exactly what the growing number of tube wells delivers — outpacing the recharge and depleting storage year on year. If recharge had exceeded withdrawal the level would rise, contradicting the data. Rainfall was normal, not absent, so a total rain failure is false. And reservoir water does not simply vanish into vapour at a rate that explains a multi-year decline driven by heavy pumping.

**51. (C)** A working recharge programme raises the water table, so the direct evidence is dried-up wells reviving and water being reached at shallower depths — the table has risen. A less salty ocean is unrelated to local groundwater recharge. Doubled monsoon rainfall would be a change in weather, not proof that recharge structures are working, and it is outside human control anyway. Rivers ceasing to flow would suggest worsening water stress, the opposite of success. Only the reviving, shallower wells show recharge succeeding.



**52. (A)** Tube wells do not make water; they only extract it from a finite aquifer recharged at a limited rate. Doubling the wells doubles the rate of withdrawal, and if that exceeds recharge the table falls and the wells eventually run dry — so the supply is not doubled permanently but undermined. The aquifer is limited, not unlimited, which is the whole reason for the flaw. And wells certainly do not add rain to the sky to refill themselves; that reverses cause and effect entirely.

**53. (C)** All four measures push the balance toward surplus: harvesting and trees raise recharge, while drip and leak repair cut withdrawal and waste, so inflow exceeds outflow and the table rises. More structures here means more water saved and recharged, not more used, so the table does not fall. None of these measures introduces salt, so the water does not turn salty. And measures that genuinely change recharge and withdrawal must move the table, so it cannot stay forever unchanged.

**54. (C)** Identical rainfall is only the inflow potential; the resulting table depends on how much each village pumps out and how much rain it actually lets soak in rather than run off — differences that can produce very different tables. Equal rain does not force equal tables, because withdrawal and infiltration vary. Concrete lowers rather than raises a table by blocking infiltration, so 'more concrete raises it' is wrong. And digging deeper merely reaches lower water; it never raises the table.

**55. (A)** Plants depend on water for their life processes, so depriving them stunts growth and cuts the harvest — a direct loss of yield. Less water does not speed growth or boost yield; that is the opposite of how plants respond to drought. Plants cannot turn absent water into vapour to thrive; you cannot transpire water you do not have. And a shortage reduces the size and health of the crop rather than altering its saltiness, so the yield falls.

**56. (D)** Sum the losses: X drops  $10+10+10 = 30$  units with a constant yearly deficit, while Y drops  $5+10+20 = 35$  units with a deficit that grows each year. So Y has fallen further and is accelerating, signalling a widening withdrawal-over-recharge gap, whereas X's gap is steady. A steady loss does not exceed Y's accelerating one here — 30 is less than 35. The totals are not equal. And Y's growing yearly figures are losses, not additions, so Y is falling, not rising.

**57. (D)** Rising vapour is just one half of a loop: it cools, condenses into clouds, and returns to the surface as rain or snow, so it is not lost but recirculated, and Earth's total water holds roughly steady as the same water changes form and location. The vapour does not escape forever, the planet is not drying out, vapour is not destroyed in the air, and it does not transmute into some new material — it remains water and comes back down, which is the essence of the water cycle.

**58. (C)** Within the water cycle the global total is essentially fixed; no rooftop device creates water. At best such a unit is rainwater harvesting — it captures rain that already fell and channels it to storage or recharge, redistributing existing water rather than making

more. It cannot manufacture new water against the cycle's conservation, it is not a desalinator turning sea breeze fresh, and it certainly does not destroy rain. Its only honest function is collecting and redirecting what is already there.

**59. (D)** World Water Day is observed on 22 March, the date set aside to focus attention on valuing and conserving fresh water. The 5 June date is World Environment Day, a broader environmental observance; 22 April is Earth Day; and 14 November is celebrated in India as Children's Day. None of these is the water-themed day, so 22 March is the correct date for World Water Day.

**60. (B)** The chapter's core message is that although Earth holds a great deal of water, the fresh water we can actually use is a tiny, scarce fraction, so it must be used wisely and managed with care. Treating water as unlimited contradicts that scarcity. Conservation is everyone's responsibility, not just farmers', since all living things depend on the shared supply. And the sea is salty and unfit for drinking or most crops, so it cannot be relied on to supply everyone's needs.